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Title: Predictive model assessment and selection in composite-based modeling using PLS-SEM: extensions and guidelines for using CVPAT

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Abstract: Purpose Researchers often stress the predictive goals of their partial least squares structural equation modeling (PLS-SEM) analyses. However, the method has long lacked a statistical test to compare different models in terms of their predictive accuracy and to establish whether a proposed model offers a significantly better out-of-sample predictive accuracy than a naïve benchmark. This paper aims to address this methodological research gap in predictive model assessment and selection in composite-based modeling. Design/methodology/approach Recent research has proposed the cross-validated predictive ability test (CVPAT) to compare theoretically established models. This paper proposes several extensions that broaden the scope of CVPAT and explains the key choices researchers must make when using them. A popular marketing model is used to illustrate the CVPAT extensions' use and to make recommendations for the interpretation and benchmarking of the results. Findings This research asserts that prediction-oriented model assessments and comparisons are essential for theory development and validation. It recommends that researchers routinely consider the application of CVPAT and its extensions when analyzing their theoretical models. Research limitations/implications The findings offer several avenues for future research to extend and strengthen prediction-oriented model assessment and comparison in PLS-SEM. Practical implications Guidelines are provided for applying CVPAT extensions and reporting the results to help researchers substantiate their models' predictive capabilities. Originality/value This research contributes to strengthening the predictive model validation practice in PLS-SEM, which is essential to derive managerial implications that are typically predictive in nature.

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